Lootex: Supporting smooth NFT trading with auto-scalable cloud computing resources

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To provide smoo
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ensure the stabili
NFT marketplace
handling instant t
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About Lootex

Founded in 2018, Lootex is a startup based in Taiwan offering an NFT trading platform dedicated for in-game digital assets. As a leading gaming NFT marketplace, Lootex currently operates more than 12,000 NFTs listed and has around 80,000 monthly active users across the world. In January 2022, the startup received $9 million in its latest seed round funding.

Industries: Technology

Google Cloud results

- Frees workforce from website performance monitoring with the autoscaling feature of GKE
- Enables software upgrades in minutes supported by GKE
- Blocks cyberattacks in four months of deployment via Google Cloud Armor

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Location: Taiwan

(https://cloud.google.com/armor/)

About Microfusion

Microfusion is a Taiwanese cloud service provider founded in 2007 focused on offering one-stop solutions and consulting with its professional architects and maintenance team to assist customers in cloud migration, IT infrastructure modernization and data analytics. Having worked with more than 2,000 customers from public and private sectors, Microfusion is a top service provider for digital transformation across various industries and was selected by CIO Magazine as one of the top 10 cloud service providers in the Asia-Pacific region in 2020.

cloud migration, IT infrastructure
aving
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Google Cloud (https://cloud.google.com/)

Cloud SQL (https://cloud.google.com/sql/)

Kubernetes Engine (https://cloud.google.com/kubernetes-engine/)

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The non-fungible token (NFT) m
an explosive boom in 2021, with
up by more than 200 times (https://www.euronews.com/next/2022/03/10/nft-sales-have-spiked-21-000-in-a-year-but-has-the-marketplace-become-a-bubble)

in just one year. Backed by blockchain technology that provides public proofs of authenticity and ownership of each token, NFTs are traded as unique digital assets that are mostly creative products like art, music and videos.

Traditionally, gamers have been trading rare items they earned or purchased in the online gaming space, but game developers have always been the real owners of these virtual assets. In NFT games, in-game items are created and sold as NFTs, which means that gamers have the full ownership of the items.

Supporters of NFT gaming believe that the adoption of NFTs will bring revolutionary changes to the industry by creating more user-centered games where gamers can enjoy a higher level of freedom. As NFTs become increasingly popular, more NFT games have been launched, and many top game developers like Ubisoft and Epic have started exploring the uses of NFTs in their games.

Believing in the potential of NFTs to liberalize gaming, Lootex (https://lootex.io/about) is dedicated to supporting the NFT gaming ecosystem by providing an NFT marketplace for in-game digital assets. Founded in 2018, the Taiwanese startup is a frontrunner in gaming NFT trading that currently operates more than 12,000 NFTs listed and has around 80,000 monthly active users across the world.

"We think introducing NFTs into online games is amazing for gamers, because it gives them more autonomy to decide how they play games and allows them to receive real rewards from playing," says David Tseng, COO at Lootex. "This is why we wanted to create a reliable marketplace for gaming NFTs."

To ensure the stability of its NFT trading platform despite the ever-increasing traffic resulting from the growing popularity of NFTs, Lootex needed to deploy its marketplace in a highly scalable cloud computing environment. In 2019, the company turned to Google Cloud (https://cloud.google.com/) for its auto-scalable computing resources and easy-to-use nature.

"Since the launch of our NFT marketplace, the incoming traffic has been growing exponentially, especially over the past year," explains Tseng. "We need a cloud platform that enables us to swiftly handle any traffic surges, so that our users can trade NFTs anytime at ease. Google Cloud fully meets our requirements with its easy-to-scale cloud infrastructure."

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Supporting smooth NFT trading under instant traffic surges with GKE

Since its adoption of Google Cloud, Lootex has adjusted the cloud architecture of its NFT marketplace several times to better process the growing amount of website traffic and transaction data. Based on Lootex's existing infrastructure, Google Cloud and partner agency Microfusion (https://www.microfusion.cloud/en/) have been providing technical advice on best practices and immediate feedback to help Lootex optimize its cloud infrastructure without disrupting its services, successfully establishing a cloud computing environment that ensures high stability and security.

"Microfusion and the Google Cloud team have been very keen to understand the issues behind our existing cloud architecture and have helped us find the best solutions," says Tseng. "Thanks to them, we're able to continuously improve the performance of our trading platform and always receive immediate technical support."

The startup now runs its NFT marketplace with virtual machines (VMs) on Google Kubernetes Engine (https://cloud.google.com/kubernetes-engine) (GKE), which can automatically scale up computing resources, distribute incoming traffic across VMs with its load balancing feature, and prevent service disruption through auto-repairing nodes. When Lootex launches an NFT project, the traffic flow of its website can instantly increase by 12 to15 times. Before using GKE, the startup needed several engineers to monitor its service and adjust VMs manually whenever a new NFT is released, which is usually during nighttime in Taiwan, because most of the users of Lootex's marketplace are based in the US and Latin America. With the auto-scaling feature of GKE, the Lootex team is able to guarantee the stability of its trading platform at any time with only one engineer responsible for monitoring.

"The automation that GKE enables has helped us provide smooth trading experiences under any kind of traffic surges, without worrying about insufficient computing resources and investing too much workforce," notes Tony Huang, tech lead at Lootex. "Since we started using GKE, we haven't experienced any downtime during high traffic due to a lack of scalability."

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Simplifying maintenance workloads and troubleshooting

For the Lootex team, another major benefit of deploying its NFT trading platform on Google Cloud is that it has simplified the maintenance workloads and troubleshooting. Every six months, the team has to upgrade the Kubernetes clusters of its VMs on GKE. Huang says that compared to other similar offerings, GKE supports the easiest and fastest way of upgrading Kubernetes clusters and enables zero downtime during the upgrade process.

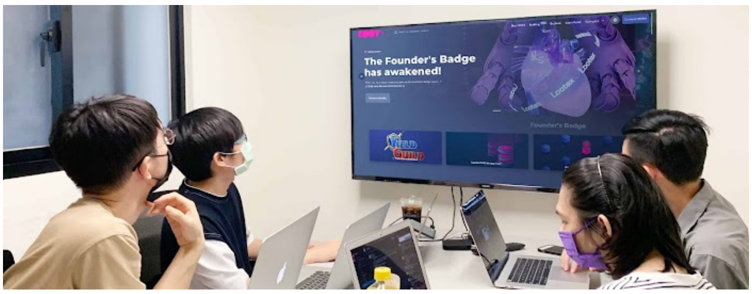
"On the other cloud platforms, it would have taken one to two work days to upgrade our Kubernetes clusters, because it requires moving clusters to a new environment," he adds. "Since Kubernetes is developed by Google, GKE is best integrated with Kubernetes and therefore allows us to upgrade our clusters in the existing environment, which only needs a few minutes."

Lootex employs Cloud SQL (https://cloud.google.com/sql) to store all the transaction data of its trading platform. Once, Lootex's platform was down because the CPUs of Cloud SQL were overloaded during an instant traffic surge. By going through the detailed log records on Cloud SQL, the Lootex team quickly identified the errors in its backend configuration that led to CPU overload, and changed the settings to accommodate any increase in the amount of data saved on Cloud SQL.

Reinforcing cybersecurity with Google Cloud Armor

Running a leading trading platform in the booming NFT industry, Lootex has been a prominent target of cyber attacks. In the past, the startup experienced DDoS attacks, which sometimes led to disruption of its trading services, or received threatening emails from hackers once a week. To strengthen the security protection of its platform, Lootex adopted Google Cloud Armor (https://cloud.google.com/armor) in early 2022, and its services have never been impacted by cyber attacks since then.

"Ensuring cybersecurity is essential for us to provide reliable and stable NFT trading services. By thoroughly detecting and mitigating web attacks, Google Cloud Armor has helped us greatly improve the quality of our services," says Tseng.



Enhancing trading platform performance for greater growth

As the NFT market keeps growing rapidly, Lootex continues to optimize the performance of its NFT marketplace to provide best-possible trading experiences. The company now uses Cloud Storage (https://cloud.google.com/storage) to support Cloud CDN (https://cloud.google.com/cdn) for faster content delivery speed. It also plans to leverage Istio (https://cloud.google.com/learn/what-is-istio) to more precisely control and distribute incoming traffic across its VMs on GKE, so that it's prepared for greater instant flow surges.

"The gaming NFT market is thriving, and we want to make sure that we have a powerful IT infrastructure to support any amount of growth in the future," notes Tseng. "With Google Cloud and the assistance of Microfusion, we've significantly improved the NFT trading experiences that we provide by minimizing the disruptions caused by traffic surges and cyber attacks. Moving forward, we're confident that we can further enhance our services through expanding our use of Google Cloud."

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